MATH1851

Notes for HKU · Spring 2024

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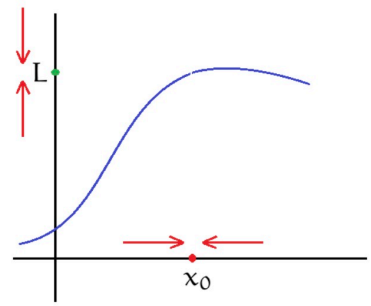
1 Limits and Continuity

1.1 Introduction to the concept of limit

We can conceptualize that the limit of a function $f(x)$ is L as x approaches c, given that we can make $f(x)$ *as close to* L *as we want* for all x sufficiently close to a, from both sides, *without actually letting* x *be* a. We can write this as:

$$\lim_{x \to a} f(x) = L$$



1.2 One-sided limits

There are two sides that x can tend to a number. We can write it as $x \to n^-$ and $x \to n^+$, which represents from the negative (left) / positive (right) side.

1.3 Existence of limits

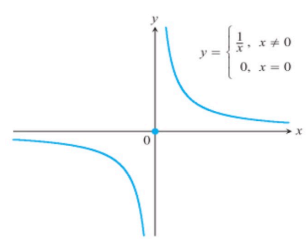
Condition for limit to exist

The limit for a function $f(x)$ only exists if and only if:

$$\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$$

WARNING: If the **limit is** ∞ it **doesn't exist**.

For this example, when $x \to 0^-, y \to -\infty$.
Similarly, as $x \to 0^+, y \to +\infty$.
Hence, we can conclude that the limit for this function as $x \to 0$ doesn't exist.



1.4 Continuity

Continuity

A function $f(x)$ is *continuous* at $x = a$ if:

$$\lim_{x \to a} f(x) = f(a)$$

Continuity properties

For f, g continuous at c, the following are also continuous at c:

1. $f \pm g$
2. kf
3. fg
4. $\frac{f}{g}$, given that $g(c) \neq 0$

Intermediate value theorem (IVT)

If a function f is continuous on $[a, b]$, there is a number c in $[a, b]$ where $f(c)$ in $[f(a), f(b)]$.

To prove that there is a root, we can use the IVT by showing there is a **change of sign** between the interval given.
To show that there's *only one solution*, check if $f'(x) > 0$ or < 0 (strict $>$) in the interval.

1.5 Computing limits

Indeterminate forms

Indeterminate forms are forms that cannot be solved by simply substituting the value of x into the function. They are:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad \infty^0, \quad 1^\infty, \quad \infty^0$$

Note that $\infty + \infty = \infty$. Related: L'Hopital's rule

Using the limit laws

For functions f, g and using $\lim\limits_{x \to a} = \bigcirc$ for simpler notation:

1. $lim_{x \to a} c = c$
2. $\bigcirc(f \pm g) = \bigcirc f \pm \bigcirc g$
3. $\bigcirc(k \cdot f) = \bigcirc k \cdot \bigcirc f$
4. $\bigcirc(f^n) = (\bigcirc f)^n$
5. $\bigcirc(fg) = \bigcirc f \bigcirc g$
6. $\bigcirc(\frac{f}{g}) = \frac{\bigcirc f}{\bigcirc g}$, given that $\bigcirc g \neq 0$. *This strict condition prevents indeterminate forms.*

We can use these laws to break a limit into separate limits, and compute that way. Also note that:

7. $\bigcirc f(g) = f(\bigcirc g)$, given that f is **continuous** at $\bigcirc g$

Limit of a polynomial

For the limit of a polynomial $p(x)$:

$$lim_{x \to a} p(x) = p(a)$$

This can be proven easily with the limit laws above.

Techniques to compute limits

To solve for limits, we have to get the expression to the right form - a polynomial, for us to substitute our limit value into the function.
To do this, often we have to **factorize** or **rationalize**.

Example 1.1 *Indeterminate forms by substitution*

This applies limit law #5. As substituting into the function directly gives $0/0$*, we have to change it into a form such that we could apply the limit laws directly.*

$$\begin{aligned} lim_{x\to 2}\frac{x^2+4x-12}{x^2-2x} &= \frac{(x-2)(x+6)}{(x-2)x} \\ &= \frac{x+6}{x} \\ \textit{Substituting 2 gives} &= 4 \end{aligned}$$

The squeeze / sandwich theorem

Suppose $f(x) \leq g(x) \leq h(x)$ in the range $[a, b]$, for c in $[a, b]$:

$$\lim_{x\to c} f \leq \lim_{x\to c} g \leq \lim_{x\to c} h$$

We will make use of the fact that the limits can be equal to solve for the limit of $g(x)$.

Example 1.2 *Squeezing a function*

When we can't seem to factorize a function, we can try squeezing it between two other functions.

$$\lim_{x\to 0} x^2 \cos\frac{1}{x}$$

We know the limits of the function $\cos\frac{1}{x}$*, so we can start from there.*

$$\textit{Given that } x \neq 0, -1 \leq \cos\frac{1}{x} \leq 1$$

$$\textit{Multiplying } x^2 \textit{ on both sides}, -x^2 \leq \cos x^2\frac{1}{x} \leq x^2$$

$$\textit{As } \lim_{x\to 0} \pm x^2 = 0, \quad \textit{we can conclude that } \lim_{x\to 0} x^2 \cos\frac{1}{x} = 0$$

1.6 Infinite limits

Determining infinite limits

If $f(x)$ gets (negatively) arbitrarily large when x approaches a, we can say:

$$\lim_{x\to a} f(x) = (-)\infty$$

After we know that the limit *may* be infinity, we then have to make sure that *the limit is the same from both sides*, so that the limit is actually ∞. We can do so by plugging numbers which are approaching the limit from both sides.

Example 1.3 *Infinite limit exists*

$$\lim_{x\to 0} \frac{6}{x^2}$$

Consider both $\lim_{x\to 0^-} \frac{6}{x^2}, \lim_{x\to 0^+} \frac{6}{x^2}$:

$$\lim_{x\to 0} \frac{6}{x^2} = \infty$$

Example 1.4 *Infinite limit doesn't exist*

$$\lim_{x\to 4} \frac{3}{(4-x)^3}$$

Checking both sides, we can conclude that the limit doesn't exist, as:

$$\lim_{x\to 4^+} \frac{3}{(4-x)^3} = -\infty, \quad \lim_{x\to 4^-} \frac{3}{(4-x)^3} = \infty$$

1.7 Limits at infinity

Infinity operations

Note the following operations:

1. $\infty + k = \infty$
2. For $k < 0,\ k\infty = -\infty$
3. $-\infty \times \infty = -\infty$

Determining limits of infinity

It is not hard to see that, for rational numbers n:

$$\lim_{x\to\pm\infty} \frac{k}{x^n} = 0$$

The easiest way to determine the limit would be to *factorize* the function so that we can use the fact above.

Determining limits of infinity of polynomials

Using the above fact, we can see that for a polynomial $p(x)$ with degree n and largest coefficient a_n:

$$\lim_{x\to\pm\infty} p(x) = a_n x^n$$

Which means we can *only consider the largest term in a polynomial* for limits of infinity.

Example 1.5 *Indeterminate forms by substitution of infinity*

Substituting ∞ into the function gives $\infty - \infty - \infty$, which is indeterminate. Hence, we must factorize it.

$$\begin{aligned}\lim_{x\to\infty} 2x^4 - x^2 - 8x &= \lim_{x\to\infty} [x^4(2 - \frac{1}{x^2} - \frac{8}{x^3})] \\ &= \infty \times 2 \\ &= \infty\end{aligned}$$

Or we can just simply use the theorem above and consider $\lim_{x\to\infty} 2x^4$ *only to give* ∞.

Example 1.6 *Factor polynomials limit to infinity*

We can simply consider the largest terms on each side and give the final answer easily.

$$\begin{aligned}
\lim_{x\to-\infty} \frac{\sqrt{3x^2+6}}{5-2x} &= \lim_{x\to-\infty} \frac{\sqrt{3x^2}}{-2x} \\
&= \lim_{x\to-\infty} \frac{\sqrt{3}|x|}{-2x} \leftarrow \sqrt{x^2} = |x| \\
&= \lim_{x\to-\infty} \frac{-\sqrt{3}}{-2} \leftarrow |c|, c < 0 = -c \\
&= \frac{\sqrt{3}}{2}
\end{aligned}$$

Note that, as we are considering the negative limit of infinity, we need to add - to the abs sign on line 3.

1.8 Asymptotes

Vertical asymptotes

f will have v-asymptotes at $x = a$ if any $\pm$ is true:

$$\lim_{x\to a^{\pm}} f(x) = \pm\infty$$

Horizontal asymptotes

f will have h-asymptotes at $y = L$ if any $\pm$ is true:

$$\lim_{x\to\pm\infty} f(x) = L$$

Related: Graphing functions

2 Trigonometry review

Trigonometric identities

1. $\sin^2 x + \cos^2 x = 1$
2. $\tan x = \frac{\sin x}{\cos x}$
3. $\csc x = \frac{1}{\sin x}$
4. $\sec x = \frac{1}{\cos x}$
5. $\cot x = \frac{1}{\tan x}$
6. $\sin 2x = 2 \sin x \cos x$
7. $\cos 2x = \cos^2 x - \sin^2 x$
8. $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$
9. $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
10. $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
11. $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
12. $1 + tan^2 x = sec^2 x$
13. $1 + cot^2 x = csc^2 x$

Related: Techniques of integration

3 Derivatives

First principle

The first principle is the definition of the derivative of a function $f(x)$ at $x = a$:

$$f'(x) = \frac{d}{dx}f(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

Note the notation of *prime* ($'$), which denotes the derivative against x for any function primed.

Differentiability

The function is differentiable at $x = a$ if:

$$\lim_{x\to a^-} f' = \lim_{x\to a^+} f'$$

Use first principle if the function is cased.

3.1 Differentiation formulas and rules

Basic formulas

- We can differentiate individual items: $(f \pm g)' = f' \pm g'$
- We can factor out a multiplicative constant: $(cf)' = cf'$
- Derivative of a constant is 0: $\frac{d}{dx}k = 0$
- Power rule: $\frac{d}{dx}x^n = nx^{n-1}$

Chain rule

Shorthand: **d**1x2 + **d**2x1

$$(u(v))' = u'(v)v' \qquad \text{or} \qquad \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Product rule

Shorthand: **d** from outside to inside

$$(uv)' = uv' + vu' \qquad \text{or} \qquad \frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Quotient rule

Shorthand: move lower **d** upper - **d** lower x upper, lower square

$$(\frac{u}{v})' = \frac{vu' - uv'}{v^2} \qquad \text{or} \qquad \frac{d}{dx}(\frac{u}{v}) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

	$f(x)$	$f'(x)$
1.	a^x	$\ln a \cdot a^x$
2.	e^{kx}	ke^{kx}
3.	$\ln kx$	x^{-1}
4.	$\lvert x\rvert$	$\frac{\lvert x\rvert}{x}$
5.	$\sin kx$	$k\cos kx$
6.	$\cos kx$	$-k\sin kx$
7.	$\tan kx$	$k\sec^2 kx$
8.	$\csc x$	$-\csc x\cot x$
9.	$\sec x$	$\sec x\tan x$
10.	$\cot x$	$-\csc^2 x$
11.	$\sin^{-1} x$	$(\sqrt{1-x^2})^{-1}$
12.	$\cos^{-1} x$	$-(\sqrt{1-x^2})^{-1}$
13.	$\tan^{-1} x$	$(1+x^2)^{-1}$

Note that the trigo derivatives can be extended to hyperbolic trigo functions, with the except of $\frac{d}{dx}\text{sech}u = -\text{sech}u\tanh u$.

3.2 Techniques of differentiation

Parametric differentiation

For a parametric equation $y = f(t)$ and $x = g(t)$:

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

Derivative of Inverse functions

At c, we first find $f'(x)$ and the value of $f^{-1}(c)$, then we apply the formula to find the value of $f^{-1'}(c)$:

$$f^{-1'}(c) = \frac{1}{f'(f^{-1}(c))}$$

$f^{-1}(c)$ can be found by solving $f(x) = c$.

Implicit differentiation

Differentiate all xy, add y' behind all differentiations of y.

To find y' for $y^2 = x^2 + \sin(xy)$:

$$\begin{aligned} y^2 &= x^2 + \sin(xy) \\ 2yy' &= \frac{d}{dx}(\sin(xy)) \\ 2yy' &= 2x + \cos(xy)(xy' + y) \end{aligned}$$

Then we simply collect terms of y'

$$y' = \frac{2x + y\cos(xy)}{2y - x\cos(xy)}$$

To find the second derivative y'', differentate the expression and substitute y' back in.

Logarithmic differentiation

For $y = f(x), \quad y' = f(x) \times (\ln f(x))'$. (Takes natural log for both sides)

To find $\frac{dy}{dx}$ for $y = x^x$:

$$\begin{aligned} y &= x^x \\ \ln y &= x \ln x \\ \frac{1}{y}\frac{dy}{dx} &= \ln x + 1 \\ \frac{dy}{dx} &= x^x(\ln x + 1) = x^x(x \ln x)' \end{aligned}$$

L'Hopital's rule

For any $a \in [\mathbb{R}, \pm\infty]$, if $\lim_{x\to a}(\frac{f}{g})$ is in indeterminate form after substitution, we can conclude:

$$\lim_{x\to a}(\frac{f}{g}) = \lim_{x\to a}(\frac{f'}{g'})$$

To find $\lim_{x\to-\infty} xe^x$, we first check if the limit is indeterminate, then we can apply the rule:

$$\begin{aligned}
\lim_{x\to-\infty} xe^x &\implies \infty \times 0 \\
\lim_{x\to-\infty} xe^x &= \lim_{x\to-\infty} \frac{x}{e^{-x}} \\
&= \lim_{x\to-\infty} \frac{1}{-e^{-x}} \text{ (rule applied)} \\
&= 0
\end{aligned}$$

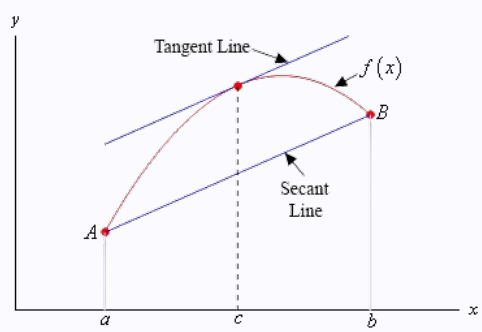
3.3 Important theorems

Mean value theorem (MVT)

For $f(x)$ that is *continuous* in $[a, b]$ and *differentiable* in (a, b):

$$f'(c) = \frac{f(b) - f(a)}{b - a}, \quad c \in (a, b)$$

The theorem tells us that, in described conditions, there must be a point c where the slope of the *tangent line* is equal to the slope of the line from $a \to b$ (secant line).

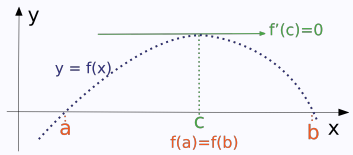


Source: https://tutorial.math.lamar.edu/Classes/CalcI/MeanValueTheorem.aspx

Rolle's theorem

For $f(x)$ that is *continuous* in $[a, b]$ and *differentiable* in (a, b):

$$\forall\ f(a) = f(b) \text{ there exists } f'(c) = 0, \quad c \in (a, b)$$



For $F' = f$, if F has 4 roots, then f has 3 roots.

3.4 Extremum points

Critical and inflection points

A **critical point** is a point where $f'(x) = 0$ or undefined, or the end-points of the domain if inclusive.
An **inflection point** is a point where $f''(x) = 0$ or undefined, that the concavity of the function changes.

Max/minimum points

The **absolute maximum/minimum points** are the points where the function has the largest/smallest value in the entire domain.
The **local maximum/minimum points** are the points where the function has the largest/smallest value in a small interval around the point.

3.5 Determining shape of graph

Concavity

Concavity is the direction of the curve, and can be described by the values of f' and f'':

f''	$-$	$+$	$+$	$-$
f'	$-$	$+$	$-$	$+$
f	↘	↙	↖	↗

Note: Arrows goes clock-wise.

	Step	Expression
1.	Determine domain of function	
2.	Special points without continuity?	
3.	Axis intercepts	$(f(x) = 0, 0)$ $(0, f(0))$
4.	Critical points	$f'(x) = 0$ or undefined
5.	Point maxima?	$+f''(x)$ or $-f''(x)$
6.	Inflection points	$f''(x) = 0$ or undefined
7.	Horizontal asymptotes	$\lim_{x\to\pm\infty} f(x) = n$?
8.	Vertical asymptotes	$\lim_{x\to a^{\pm}} f(x) = \pm\infty$?
9.	Area strictly increasing/decreasing?	$f'(x) > 0$ or $f'(x) < 0$
10.	Area concavity?	$- + + - / - + - +$/

Related: Definition of asymptotes

Higher derivatives

Derivatives of higher order (e.g. $f''(x), f'''(x)$) can be expressed as $f^{(n)}(x)$

4 Integrals

Definition of natural logs

$$\ln x = \int_1^x \frac{1}{t} dt, \quad x > 0$$

4.1 Definite integrals

Signed areas

Signed area is the area between the curve and the x-axis, where the area above the x-axis is positive and below is negative.

Properties of definite intergrals

1. $\int_a^b f(x)dx = -\int_b^a f(x)dx$
2. $\int_a^a f(x)dx = 0$
3. $\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$
4. $\int_a^b kf(x)dx = k\int_a^b f(x)dx$
5. $\int_a^b (f(x) \pm g(x))dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$

4.2 Fundamental theorem of calculus

Fundamental theorem of calculus (FTC)

If $f(x)$ is continuous in $[a, b]$, then:

$$\int_a^b f(x)dx = F(b) - F(a)$$

Where $F(x)$ is the definite integral of $f(x)$.

Second fundamental theorem of calculus

If $f(x)$ is continuous in $[a, b]$, then:

$$\frac{d}{dx}\int_a^x f(t)dt = f(x)$$

Find $g'(x)$ for $g(x) = \int_1^{x^2} \cos t \; dt$:
We first define $G(u) = \int_1^u \cos t \; dt$, then we can apply the chain rule:

$$\begin{aligned} g'(x) &= (G(x^2))' \\ &= G'(x^2) \cdot 2x \\ &= \cos x^2 \cdot 2x \end{aligned}$$

Cases of area 0

For $\int_a^b f(x)dx$:

- If $b = a$, then the area is 0 (no width)
- If $f(x) = f(-x)$ and $a = -b$, then the area is 0 (symmetry)

4.3 Techniques of integration

Overview

- By **substitution**: "sub $g(x) \to u$, find du and replace all functions of $x(dx)$ with $u(du)$"
- By **part**: "$\int ab = a \int b - \int (a' \times \int b)$"
- By **joining recurring parts**: "$\int ab = f(ab) + \int ab \to 2 \int ab = f(ab)$"
- By **partial fractions**: "$\frac{k}{f(x)g(x)} = \frac{A}{f(x)} + \frac{B}{g(x)}$"
- By **trigonometric identities**
- By **multiplying fractions by one**: "Multiplying $\frac{\sec^2 x}{\sec^2 x}$ to fit trigo form"

Integration by substitution

For $\int f(g)g' \, dx$:

1. Let $u = g$
2. Find $du = g'dx$
3. Change limits in terms of u if definite
4. Replace all $g \to u$ and $g'dx \to du$
5. Integrate, then subsitute $u \to g$

Integration by parts

$$\int ab = a \int b - \int (a' \times \int b)$$

Tips: Let a (differentiating term) to the first term you see on the list:

1. **L**ogarithmic
2. **I**nverse trigo
3. **A**lebratic (polynomial)
4. **T**rigo
5. **E**xponential

Deriving the IBP formula

The formula is derived from the product rule of differentiation:

$$
\begin{aligned}
(ab)' &= a'b + b'a \\
b'a &= (ab)' - a'b \\
\int b'a &= ab - \int a'b \\
\int ab &= a\int b - \int (a' \times \int b)
\end{aligned}
$$

Trigo-identities substitution tips

Use the following trigonometric identities to simplify the integral:

- $\sin^2 x + \cos^2 x = 1$
- $\sin^2 x = \frac{1-\cos 2x}{2}$
- $\cos^2 x = \frac{1+\cos 2x}{2}$
- $2\sin x \cos x = \sin 2x$
- $\cos 2x = \cos^2 x - \sin^2 x$
- $\sec^2 x = 1 + \tan^2 x$
- $\csc^2 x = 1 + \cot^2 x$

Use the following substitutions for similar expressions:

$$
\text{Expressions of the form } \begin{cases} a^2 - f(x)^2 & \to f(x) = a\sin\theta \\ a^2 + f(x)^2 & \to f(x) = a\tan\theta \\ f(x)^2 - a^2 & \to f(x) = a\sec\theta \end{cases}
$$

Remember that this uses *integration by substitution*, so we need to find dx in terms of θ.

4.4 Solids of revolution

Solids of revolution

A 3D shape formed by rotating a *region* around an axis.

A region is defined by 2 curves and an interval:

1. The vertical region bounded by an outer cruve $R(x)$ and an inner curve $r(x)$.
2. The inner curve is $y = 0$ for the region bounded by the x-axis.
3. The horizontal region is bounded by an interval $x: \ [a, b]$

Note that for the methods below, the terms x and y can be *switched* to fit the problem.
$\{f(x) \to \circlearrowright x\}$ denotes using a function of x to find the volume of a solid of revolution around the x-axis.
If the volume is generated by the rotation about an axis other than the main axes, we can simply *shift* the function to fit the main axes.

Volume by Washers (Disk)

$$\{f(x) \to \circlearrowright x\}: \quad V = \pi \int_a^b R^2(x) - r^2(x)dx$$

For rotating about $y = n$: $\forall R/r(x) \to n - R/r(x)$

Volume by Cylindrical shells

$$\{f(x) \to \circlearrowleft y\}: \quad V = 2\pi \int_a^b [x][R(x) - r(x)]dx$$

For rotating about $x = n$: $[x] \to [x - n]$

4.5 Arcs and surfaces

Arc length

The *arc length* of a curve $y = f(x)$ from $x = a$ to $x = b$ is given by:

$$L = \int_a^b \sqrt{1 + f'^2(x)}dx$$

And for parametized equations $x = f(t)$, $y = g(t)$:

$$L = \int_a^b \sqrt{f'^2(t) + g'^2(t)}dt$$

Area surfaces of revolution

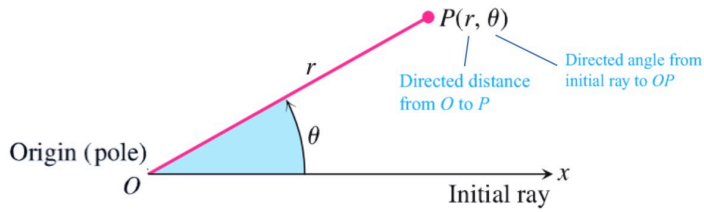
Used to find the *surface area* generated by rotating a curve along an axis.

$$\{f(x) \to \circlearrowright x\}: \quad A = 2\pi \int_a^b f(x)\sqrt{1 + f'^2(x)}dx$$

And for parametized equations $x = f(t)$, $y = g(t)$:

$$A = 2\pi \int_a^b g(t)\sqrt{f'^2(t) + g'^2(t)}dt$$

5 Polar Coordinates



Equations relating Rolar and Cartesian coordinates

We can convert between polar and Cartesian coordinates using:

$$x = r\cos\theta, \quad y = r\sin\theta$$

We can also see that:

$$r^2 = x^2 + y^2. \quad \tan\theta = \frac{y}{x}$$

The polar equation for the circle $r^2 = x^2 + y^2$ is simply $r = a$.

Slope of polar curves

The slope of a polar curve at a point is given by:

$$\frac{dy}{dx} = \frac{r\cos\theta + r'\sin\theta}{-r\sin\theta + r'\cos\theta}$$

To the the slope of the tangent to the unit circle at Cartesian coordinates $(\frac{1}{2}, \frac{\sqrt{3}}{2})$:
We can find the the polar coordinates to be $(1, \frac{\pi}{3})$ using $\cos^{-1}\frac{1}{2} = \frac{\pi}{3}$, then we can use the formula to find the slope:

$$\frac{dy}{dx} = \frac{1\cos\frac{\pi}{3} + 1'\sin\frac{\pi}{3}}{-1\sin\frac{\pi}{3} + 1'\cos\frac{\pi}{3}} = -\frac{1}{\sqrt{3}}$$

Graphing

We can easily graph any polar formula $r = f(\theta)$ following the following steps:

1. Plot $r - \theta$ graph. If the function involves trigo functions, we can simply translate its graph. The height at a certain point is the distance from the origin.
2. Then we can simply trace the graph from the left, and plot the points according to the distance from the origin at each angle.
3. If the graph is symmetric, we can simply copy the other half.

6 First Order Differential Equations

Differential equations are equations that involve a function and its derivatives.

Order of differential equations

A n ordered differential equation is an equation of the form:

$$F(x, y, y' \dots y'_n) = 0$$

Where y'_n is the nth derivative of y with respect to x. The highest degree of the derivative is n for a n-ordered differential equation. Note that y is really just $y(x)$ (A function of x)

6.1 Solving linear 1st-ODEs

Linear differential equations

Linear differential equations does not contain non-linear functions. (e.g. $\sin y$) Otherwise, it's a non-linear ODE.

Solving by integrating factors

We can solve a linear 1st-ODE as followed, given a **particular solution** of $y(x)$:

$$y' + p(x)y = q(x) \quad : \quad \times\ e^{\int p(x)}$$

The multiplied integration factor $e^{p(x)}$ will give us a product of the *product rule*, then we simply integrate both sides to solve for y. Make sure that the **coefficient of y' is 1**.

Separable equations

We can solve a separable equation as followed, given a **particular solution** of $y(x)$:

$$\begin{aligned} \frac{dy}{dx} &= f(x)g(y) \\ g(y)^{-1}dy &= f(x)dx \\ \int g(y)^{-1}\, dy &= \int f(x)\, dx \Leftarrow [(x, y) \to c] \end{aligned}$$

6.2 Solving non-linear 1st-ODEs

Bernoulli's equation

A non-linear 1st-ODE of the form can be solved by:

$$y' + p(x)y = q(x)y^n \quad n \in \mathbb{R} \quad : \quad \text{sub } u = y^{1-n} \to y, y'$$

The subsitution $u = y^{1-n}$ will turn the equation into a linear ODE, then simply solve using integrating factors.

Riccati's equation

A non-linear 1st-ODE of the form can be solved by the following, given a **particular solution** of $y(x)$:

$$y' = p(x)y^2 + q(x)y + r(x) \quad : \quad \text{sub } y = y(x) + u^{-1}$$

Homogeneous equations

A homogeneous equation has it's x and y terms **in the same degree**. (e.g. $x^2 + xy + y^2 = 0$)
A *homogeneous* 1st-ODE of the form can be solved by the following, given a **particular solution** of $y(x)$:

$$y' = f(\frac{y}{x}) \quad : \quad \text{sub } u = \frac{y}{x} \to y' = u + xu'$$

We can **divide the formula** by x^n or y^n to get the equation in the desired form (*every term is the ratio* $\frac{y}{x}$). Otherwise, we can shift the origin using $X = x - n$ and $Y = y - m$.
After substitution, we will get a **separable equation** after the substitution, and the particular solution is used.

6.3 Exact equations

Partial derivatives

A partial derivative is a derivative of a function with respect to one of its variables, with the others held constant. The following notation expresses the partial derivative of f with respect to x:

$$\frac{\partial f}{\partial x}$$

$$F = 2x + y$$
$$\frac{\partial F}{\partial x} = 2$$

Exact equations

An exact equation is simply a 1st-ODE where $dF = 0$.
The expressed equation dF is exact if:

$$dF = Mdx + Ndy \quad : \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solving exact equations

To find the solution of an exact equation:

$$M = \frac{\partial F}{\partial x}, \quad N = \frac{\partial F}{\partial y}$$

$$\partial F = M \partial x$$

$$F = \int M dx + g(y)$$

$$\frac{\partial F}{\partial y} = N$$

$$\frac{\partial \int M dx + g(y)}{\partial y} = N \to g'(y)$$

$$\int g'(y)\, dy = g(y) \to F$$

$g(y)$ is present as we are integrating partially with respect to x, and $g(y)$ is the constant of integration. Hence, the solution would be:

$$\int M dx + g(y) = c$$

7 Second Order Differential Equations

7.1 Solving homogeneous linear 2nd-ODEs

Constant coefficient Homogeneous 2nd-ODEs

The term homogeneous is used differently from the previous section.
A homogeneous 2nd-ODE is of the form:

$$ay'' + by' + cy = \mathbf{0}$$

Where a, b, c are constants. (Coefficients are *constants*)

We first use the following substitution:

$$ay'' + by' + cy = 0 : \quad y = e^{\lambda x} \implies a\lambda^2 + b\lambda + c = 0 \to \lambda$$

To find the general solution, we put the λ roots into the **quadratic characteristic equation**:

1. $\lambda_1 \neq \lambda_2 : \quad y = c_1\mathbf{e}^{\lambda_1 x} + c_2\mathbf{e}^{\lambda_2 x}$
2. $\lambda_1 = \lambda_2 : \quad y = c_1\mathbf{e}^{\lambda x} + c_2 x\mathbf{e}^{\lambda x}$
3. $\{\lambda_{1,2} = \alpha \pm \beta i\} \in \mathbb{C} : \quad y = c_1\mathbf{e}^{\alpha x}\cos(\beta x) + c_2\mathbf{e}^{\alpha x}\sin(\beta x)$

If given **particular solutions** of y and y', we can solve for c_1 and c_2 by finding y' with our general solution and substituting.

Cauchy-Euler equations

A Cauchy-Euler equation is a slight variation of homogenous 2nd-ODEs, which is of the following form and can be solved by:

$$ax^2y'' + bxy' + cy = 0 \quad : \quad y = x^\lambda \implies a(\lambda^2 - \lambda) + b\lambda + c = 0$$

The general solutions is similar to that of the homogeneous 2nd-ODEs, but with **all terms of** $x \to \ln x$:

1. $e^{\lambda x} \to x^\lambda$
2. $e^{\lambda x} \to x^\lambda, \quad x \to \ln x$
3. $e^{\alpha x} \to x^\alpha, \quad \beta x \to \beta \ln x$

7.2 Solving non-homogeneous linear 2nd-ODEs

Constant coefficient Non-homogeneous 2nd-ODEs

A non-homogeneous 2nd-ODE is of the form:

$$F: \ ay'' + by' + cy = g(x)$$

Where a, b, c are constants. (Coefficients are *constants*)

We first solve for $\lambda_{1,2}$ for the **complementary homogenous function** F_c to get Y_c:

$$F_c: \ ay'' + by' + cy = 0 \ \rightarrow Y_c$$

The general solution y for the non-homogeneous 2nd-ODE F is:

$$y = Y_c + Y_p$$

Where Y_p is a **particular solution** of y. To solve for Y_p, we can use the following methods:

1. **Method of undetermined coefficients**
2. **Variation of parameters**

Method of undetermined coefficients

To solve for Y_p for a non-homogeneous 2nd-ODE, let Y_p as the following if $g(x)$ consists of:

- $e^{ax} \rightarrow Y_p = Ae^{ax}$
- $\sin x$ and / or $\cos x \rightarrow Y_p = A\sin x + B\cos x$
- $x^n \rightarrow Y_p = Ax^n = A_n x^n + A_{n-1}x^{n-1} + \cdots + A_0$ (polynomial of degree n)

Important things to note:

▶ If $g(x)$ is a product of multiple components, Y_p **is the product of the different results**.

▶ If Y_p consists of a non-constant **term that exists in** Y_c, we **must multiply** Y_p by x^i and repeat the process.

We then substitute $y = Y_p \rightarrow F$ and solve for A and B.

For Cachy-Eular equations, we instead multiply Y_p by $(\ln x)$ if Y_p consists of a non-constant term that exists in Y_c.

Variation of parameters

We can use this method when we are unable to see a particular solution for Y_p in the above method. Note that for Y_c is in the form of $c_1y_1 + c_2y_2$. To solve for Y_p for a non-homogeneous 2nd-ODE:

$$Y_P = -y_1 \int \frac{y_2 g(x)}{W} dx + y_2 \int \frac{y_1 g(x)}{W} dx, \quad W = y_1 y_2' - y_2 y_1'$$

Note that for Cachy-Eular equations, $g(x)$ is defined as the function with the coefficient of y'' as 1, hence, $g(x) \rightarrow \frac{g(x)}{ax^2}$.

8 Solving ODEs with Laplace Transforms

Laplace transform

The Laplace transform is a technique used to solve linear ODEs with constant coefficients. The Laplace transform of a function $f(t)$ is defined as:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)dt = F(t)$$

Where s is a complex number.

Properties of Laplace transforms

The following are some properties of Laplace transforms:

- $\mathcal{L}\{f+g\} = \mathcal{L}\{f\} + \mathcal{L}\{g\}$
- $\mathcal{L}\{kf\} = k\mathcal{L}\{f\}$

Laplace transform of derivatives

The Laplace transform of the derivative of a function $y(t)$ is:

$$\mathcal{L}\{y'\} = sY(s) - y(0)$$
$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$$

Where $f(0)$ is the initial condition of f.

	f	$\mathcal{L}\{f\}$
0.	a	$\frac{a}{s}$
0.	e^{at}	$\frac{1}{s-a}$
0.	$f * g$	$F(s)G(s)$
1.	t^n	$\frac{n!}{s^{n+1}}$
2.	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n}\mathcal{L}\{f(t)\}$
3.	$\sin at$	$\frac{a}{s^2+a^2}$
4.	$\cos at$	$\frac{s}{s^2+a^2}$
5.	$e^{at}f(t)$	$F(s-a)$
6.	$f(t-a)H(t-a)$	$e^{-as}\mathcal{L}\{f(t)\}$

Convolution operator: $f * g(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau$

Inverse Laplace transform

The inverse Laplace transform is the reverse operation of the Laplace transform.

$$\mathcal{L}^{-1}\{F(t)\} = f(t)$$

You basically think backwards like how you'd do intergration sometimes. Remember to use the rules!

Partial fractions

A fraction can be decomposed into partial fractions if the degree of the numerator is **less than the degree of the denominator**. The following are the types of partial fractions:

1. $\frac{f(x)}{g(x)h(x)} = \frac{A}{g(x)} + \frac{B}{h(x)}$
2. $\frac{f(x)}{g^2(x)h(x)} = \frac{A}{g(x)} + \frac{B}{g^2(x)} + \frac{C}{h(x)}$

We can solve for A, B, and C by multiplying the denominator (to make left side $f(x)$ only) and solving for the numerator.

Solving ODEs with Laplace transforms

To solve a linear ODE with constant coefficients using Laplace transforms:

1. Take the Laplace transform of both sides of ODE ($y(t) \to Y(s)$)
2. Solve for the Laplace transform of the function $Y(s)$
3. Convert $Y(s)$ into partial fractions
4. Find the inverse Laplace transform of the function ($Y(s) \to y(t)$)

9 Summary of formulas

Concept	**Formula**
Limit exists	$\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x)$
Is continuous	$f(a) = \lim_{x\to a} f(x)$
Is differentiable	$\lim_{x\to a^-} f' = \lim_{x\to a^+} f'$ (for cased, use first principle)
IVT	If f continuous in $[a,b]$, then $\exists\ c \in [a,b] : f(c) \in [f(a), f(b)]$
Continuous limit law	$\lim_{x\to a} f(g(x)) = f(\lim_{x\to a} g(x))$
Polynomial limit law	$\lim_{x\to a} p(x) = p(a)$
Sandwich theorem	$f(x) \le g(x) \le h(x) \to \lim_{x\to a} g(x) = L$
∞ limit exists	$\lim_{x\to a^\pm} f(x) = \infty$
First principle	$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$
Chain rule	$(u(v))' = u'(v)v'$
Product rule	$(uv)' = u'v + uv'$
Quotient rule	$\left(\frac{u}{v}\right)' = \frac{u'v-uv'}{v^2}$
Inverse derivatives	$f^{-1'}(c) = \frac{1}{f'(f^{-1}(c))}$
Implicit differentiation	$(y^2)' = 2yy'$
Logarithmic differentiation	Take nautral log of both sides
L'Hopital's rule	$\lim_{x\to a} \frac{f(x)}{g(x)}$: indeterminate $\implies \lim_{x\to a} \frac{f'(x)}{g'(x)}$
MVT	$c \in (a,b) : f'(c) = \frac{f(b)-f(a)}{b-a}$ given conditions
Rolle's theorem	$c \in (a,b) : f'(c) = 0$ given conditions
Axis intercepts	$x : \{f(x) = 0, 0\}, \quad y : \{0, f(0)\}$
Critical points	$f'(x) = 0$ or undefined
Inflection points	$f''(x) = 0$ or undefined
Vertical asymptotes	$\lim_{x\to a^\pm} f(x) = \pm\infty$
Horizontal asymptotes	$\lim_{x\to\pm\infty} f(x) = L$
Area strictly increasing/decreasing	$f'(x) > 0$ or $f'(x) < 0$
Concavity	$f'' - + + - \ / \ f' - + - + \ / \ f$ ↷ ↶ ↰ ↱
FTC	$\int_a^b f(x)dx = F(b) - F(a)$
Second FTC	$\frac{d}{dx}\int_a^x f(t)dt = f(x)$
Integration by substitution	sub $g(x) \to u$, find du, $x(dx) \to u(du)$
Integration by parts	$\int ab = a\int b - \int(a' \times \int b)$, let diff term be first in $[L, I, A, T, E]$
Joining recurring parts	$\int ab = f(ab) + \int ab \to 2\int ab = f(ab)$
Partial fractions	$\frac{k}{f(x)g(x)} = \frac{A}{f(x)} + \frac{B}{g(x)}$
Trigo substitution	$\begin{cases} a^2 - f(x)^2 & \to f(x) = a\sin\theta \\ a^2 + f(x)^2 & \to f(x) = a\tan\theta \\ f(x)^2 - a^2 & \to f(x) = a\sec\theta \end{cases}$
Volume by Washers (Disk)	$\{f(x) \to \circlearrowright x\} : \quad V = \pi\int_a^b R^2(x) - r^2(x)dx$
Volume by Cylindrical shells	$\{f(x) \to \circlearrowright y\} : \quad V = 2\pi\int_a^b [x][R(x) - r(x)]dx$
Arc length	$L = \int_a^b \sqrt{1 + f'^2(x)}dx$
Area surfaces of revolution	$\{f(x) \to \circlearrowright x\} : \quad A = 2\pi\int_a^b f(x)\sqrt{1 + f'^2(x)}dx$
Parametized equations (above 2)	$\sqrt{f'^2(t) + g'^2(t)}dt$

Concept	**Formula**
ODE Integrating factors	$y' + p(x)y = q(x) \quad : \quad \times\ e^{\int p(x)}$
ODE Separable	$\frac{dy}{dx} = f(x)g(y) \quad : \quad g(y)^{-1}dy = f(x)dx$
ODE Bernoulli's	$y' + p(x)y = q(x)y^n \quad n \in \mathbb{R} \quad : \quad \text{sub } u = y^{1-n} \to y, y'$
ODE Riccati's	$y' = p(x)y^2 + q(x)y + r(x) \quad : \quad \text{sub } y = y(x) + u^{-1}$
ODE Homogeneous	$y' = f(\frac{y}{x}) \quad : \quad \text{sub } u = \frac{y}{x} \to y' = u + xu'$
Exact equations	$dF = Mdx + Ndy \quad : \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
Solving exact equations	$M = \frac{\partial F}{\partial x}, \quad N = \frac{\partial F}{\partial y} \quad : \quad \int Mdx + g(y) = c$
2ODE Homogeneous	$ay'' + by' + cy = 0 \quad : \quad y = e^{\lambda x} \implies a\lambda^2 + b\lambda + c = 0 \to \lambda$
2ODE Cauchy-Euler	$ax^2y'' + bxy' + cy = 0 \quad : \quad y = x^\lambda \implies a(\lambda^2 - \lambda) + b\lambda + c = 0$
2ODE Non-homogeneous	$ay'' + by' + cy = g(x) \quad : \quad y = Y_c + Y_p$
Characteristic solution	$Y_c = \begin{cases} \lambda_1 \neq \lambda_2 : & c_1\mathbf{e}^{\lambda_1 x} + c_2\mathbf{e}^{\lambda_2 x} \\ \lambda_1 = \lambda_2 : & c_1\mathbf{e}^{\lambda x} + c_2 x\mathbf{e}^{\lambda x} \\ \{\lambda_{1,2} = \alpha \pm \beta i\} \in \mathbb{C} : & c_1\mathbf{e}^{\alpha x}\cos(\beta x) + c_2\mathbf{e}^{\alpha x}\sin(\beta x) \end{cases}$
Method of undetermined	$Y_p = \begin{cases} e^{ax} : & Ae^{ax} \\ \sin x \text{ and / or } \cos x : & A\sin x + B\cos x \\ x^n : & \text{n degree polynomial} \end{cases}$
Variation of parameters	$Y_P = -y_1 \int \frac{y_2 g(x)}{W}dx + y_2 \int \frac{y_1 g(x)}{W}dx, \quad W = y_1y_2' - y_2y_1', Y_c = c_1y_1 + c_2y_2$
Cauchy Y_c & Y_p	$x \to \ln x$
Laplace derivatives	$\mathcal{L}\{y'\} = sY(s) - y(0), \quad \mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0), \quad \mathcal{L}\{f * f\} = FF$
Laplace rules (not given)	$\mathcal{L}\{a\} = \frac{a}{s}, \quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$
Solve by Laplace	$y(t) \to Y(s) \to$ change subject $Y(s) \to y(t)$